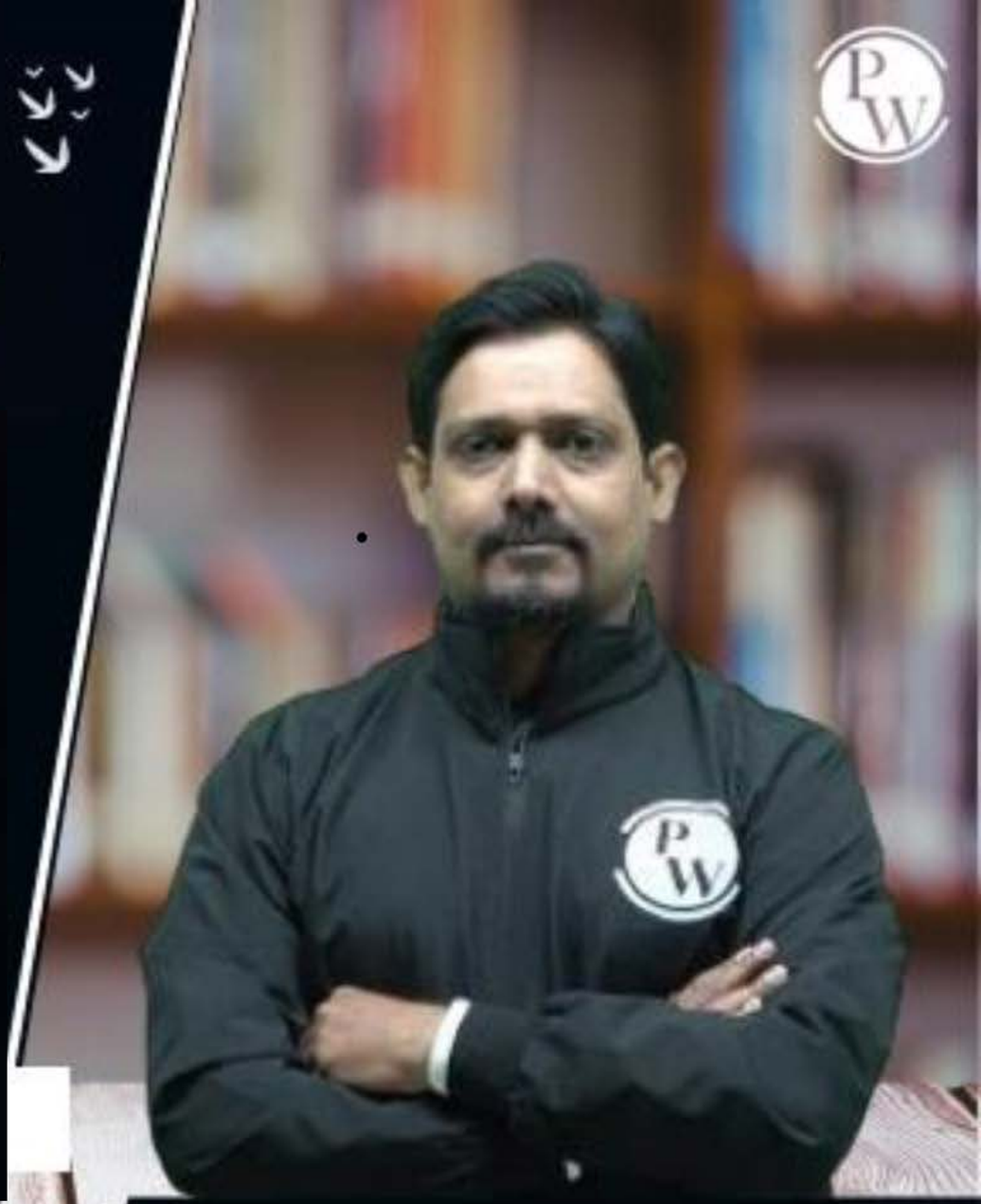
Computer Science & IT

Data Structure & Programming

Array

Lecture No. 01



By- Abhishek Sir

Topics to be Covered



- Topic — Syllabus
- Topic — Books
- Topic — Weightage
- Topic
- Topic

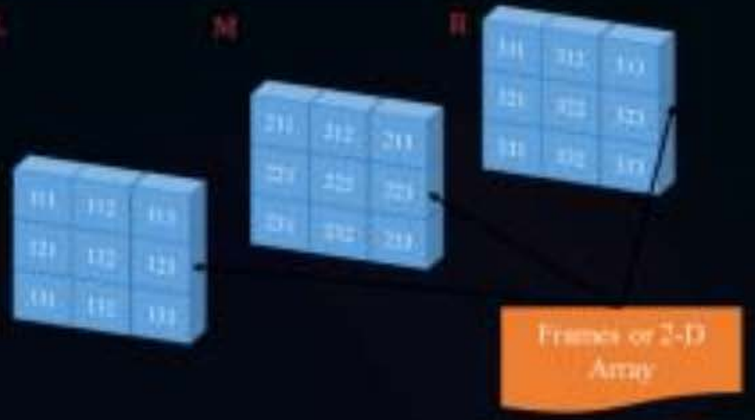
56 Hrs

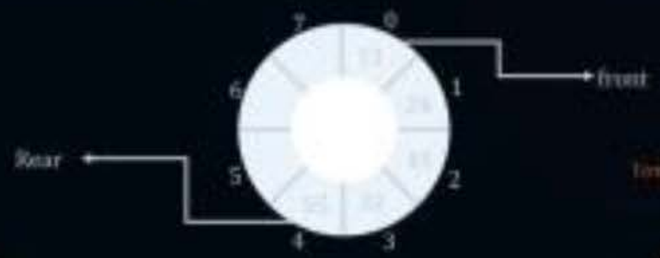
28 days

Slide

Topic : Data Structure

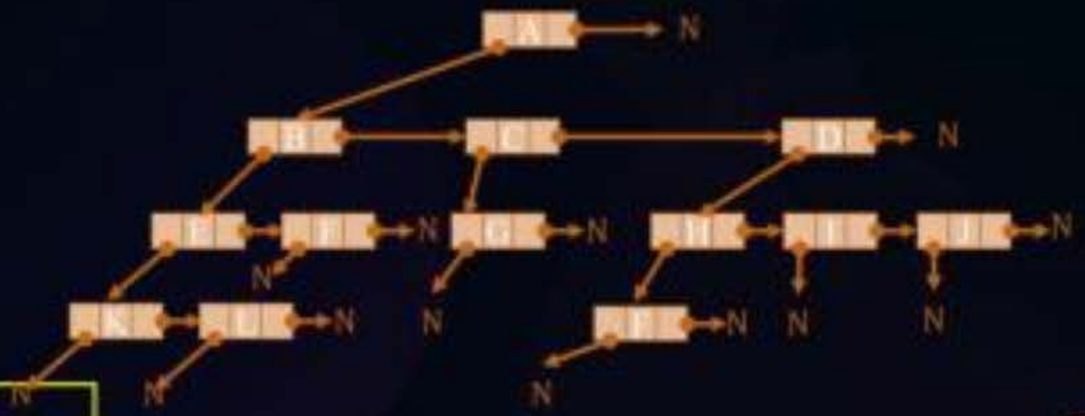


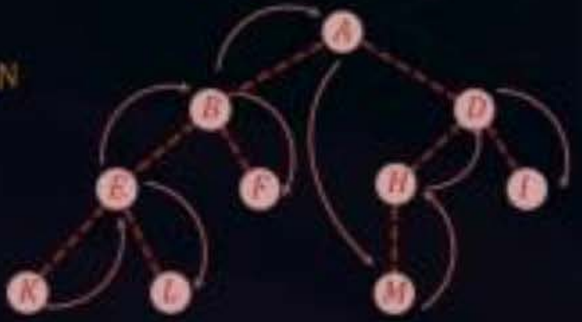














Slide



Topic : Syllabus

programming in 'C'. Array, Linked List, stack, queue

Tree, Binay tree, Binay Search tree (AVL tree)

Binary Heap, Graph (Representation)

Slide



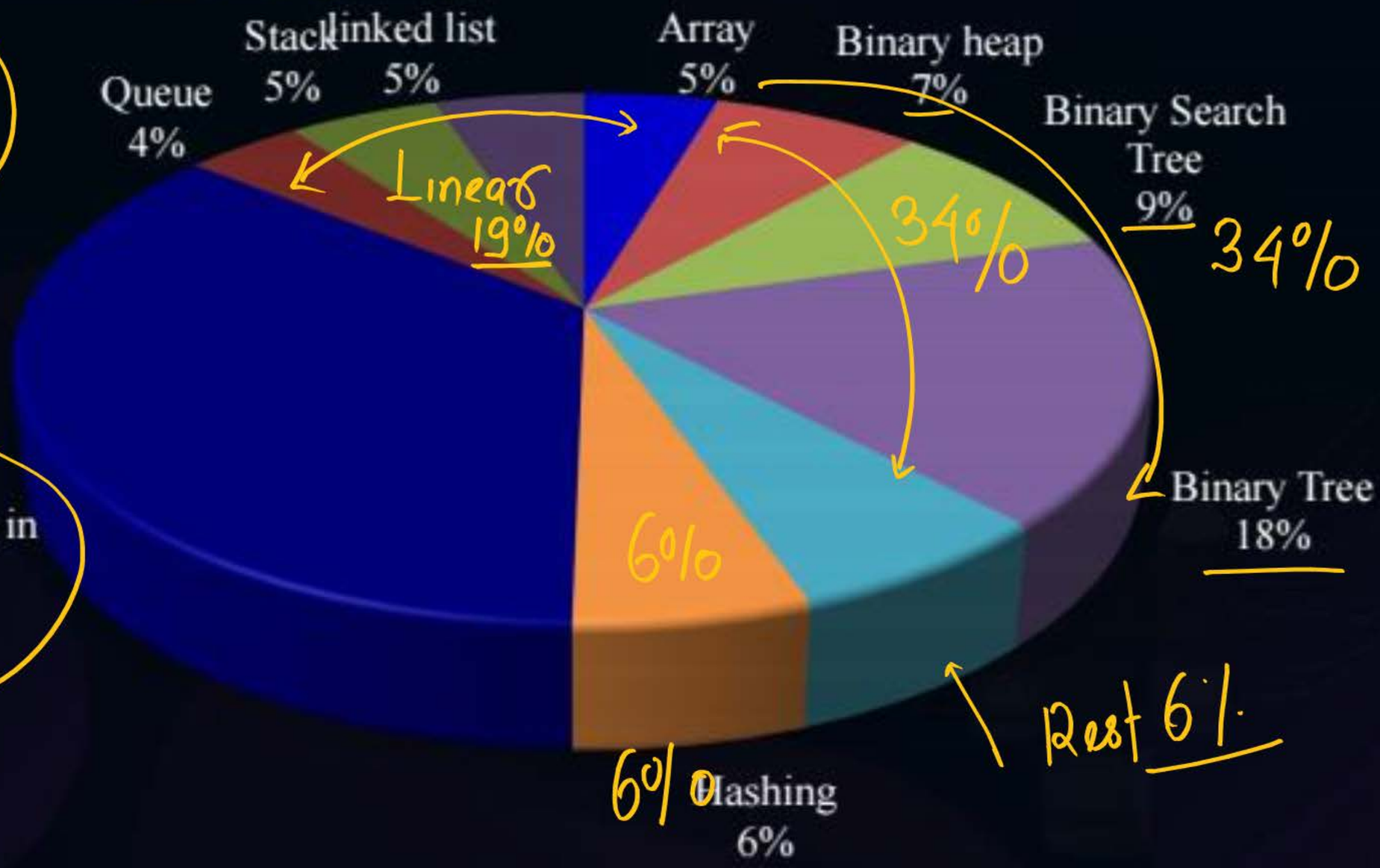
Topic : Syllabus

Syllabus

- programming in 'c' — 35%
- Linear Data Structure — 19%
  - Array, stack, queue
  - Linked List
- Non Linear Data Structure — 34%
  - * Tree
  - * Binary Search Tree
  - * AVL tree
  - * Binary Heap
  - * Graph
  - * Hash table

Slide

Topic : Weightage



Slide

Expected qushon 5-6 Qustion

Expecked marks - 8-10 marks



Topic : Definition

Collection of Data

Data structure is about data organization and management, storage format that leads to efficient access & modification (operation performed on them)

Slide



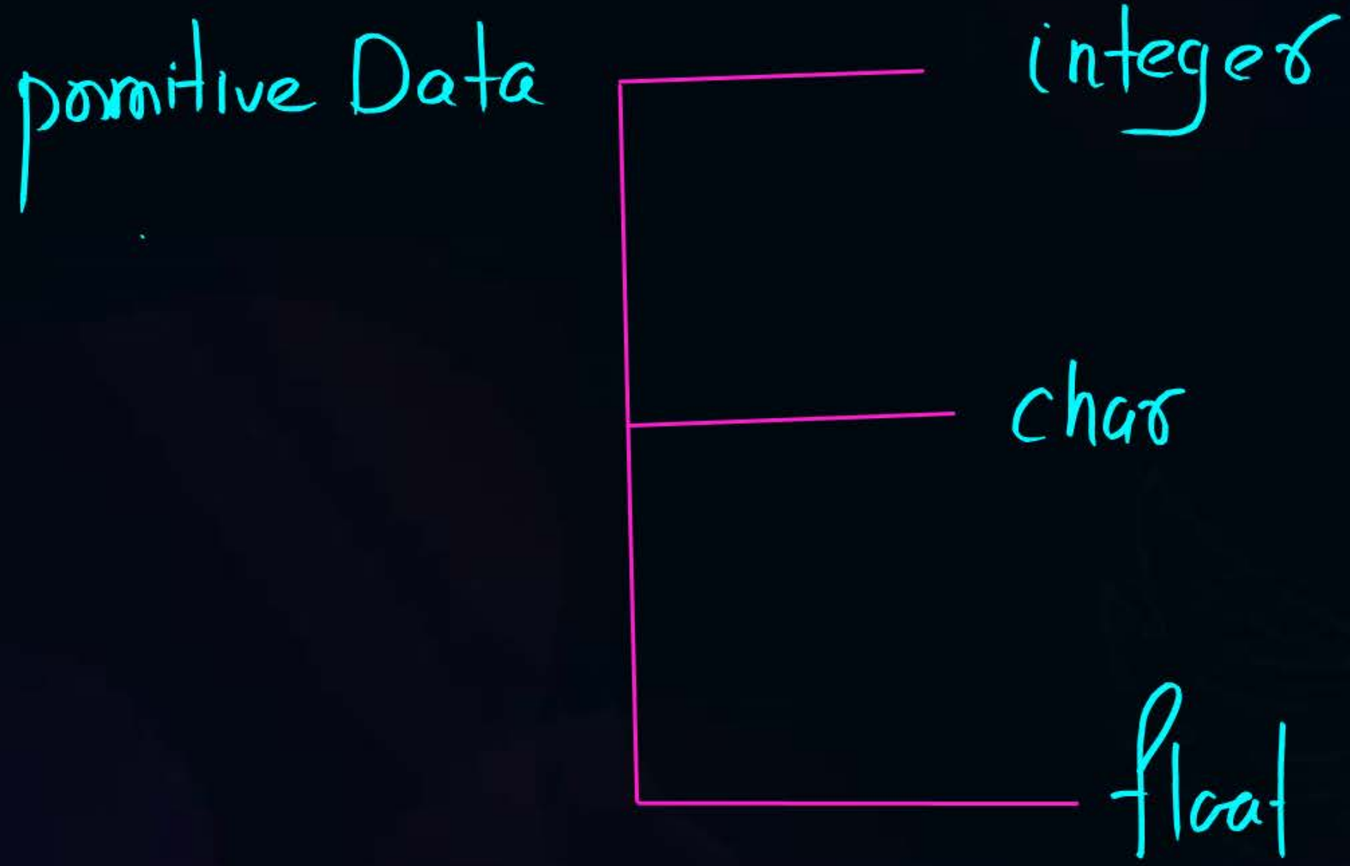
Topic : Classification of Data Structures



Slide



Topic : Classification of Data Structures



Slide



Topic : Classification of Data Structures

Abstract Data Type

Abstract data types define Data & its corresponding operation without implementation.

Stack : Linear collection of element based on LIFO property

main opn ① push : push element on top of stack.

pop removes element from top of stack

Slide



Topic : Classification of Data Structures

Linear Search

if we want to sear

Unordered array : 11, 1, 23, 16, 2, 5

ordered array : Increasing order →
20, 26, 29, 30, 35

: 98 64, 25, 12, 11
Decreasing order

Slide



Topic : Classification of Data Structures

ordered :

4. 14, 14, 15, 16, 17, 17, 18

Non Decreasing order

14 → 14

10 9, 9, 8, 7, 7, 6, 5

Non Increasing order.

Slide



Topic : Classification of Data Structures

* if we want to Search an element in unordered List then we Linear Search.

* if we want to search an element ordered list then we binary Search

Slide



Topic : Classification of Data Structures

18	1	98	22	48	56	12	9	10
1	2	3	4	5	6	7	8	9

Unordered

13=18 13=1 13=98

Suppose we want to search (13)

Suppose we search 18 in this collection 1 Comparison

the 13 compared with 1st position element
2nd position element
.
.
.
9th position element

Slide



ONE after another element Searching is Linear Search

Suppose I am searching X

Maximum No. of comparison ; Collection Size = 9

Collection of element size : N

Maximum Comparison N



Topic : Dynamic Data Structure

Complexity Theory Maximum O(n)

order

n is size of array

Maximum value

Notation

Slide



Topic : Abstract Data Types

Maximum (collection of element)

↓

O(n) ← Algorithm

Slide



Topic : Abstract Data Types

CS

Minimum No. of comparison

$\Omega(1)$

Minimum

No. of comparison for any value of n

10 — 1 comparison

1000 — 1 comparison

10^6 — 1 comparison

Slide



Topic : One

Array

Address

Collection of element

Size

```
int LinearSearch(int a[], int x, int n){
    int i;
    for(i=0; i<n; i++){
        if(a[i]==x)
            return i;
    }
    return -1;
}
```

element to be searched

Comparison

terminate

H.W

understand

↑

Telegram

Slide



2 mins Summary

Topic Weightage

Topic Syllabus

Topic Classification

Topic Maximum - 0

Topic Minimum

Slide

THANK - YOU